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#include <stdio.h>

int main() {
    int n, m, i, j, k;
    printf("Enter number of processes: ");
    scanf("%d", &n);

    printf("Enter number of resources: ");
    scanf("%d", &m);

    int alloc[n][m], max[n][m], need[n][m];
    int avail[m];

    printf("Enter Allocation Matrix:\n");
    for(i = 0; i < n; i++) {
        for(j = 0; j < m; j++) {
            scanf("%d", &alloc[i][j]);
        }
    }

    printf("Enter Maximum Demand Matrix:\n");
    for(i = 0; i < n; i++) {
        for(j = 0; j < m; j++) {
            scanf("%d", &max[i][j]);
        }
    }

    printf("Enter Available Resources:\n");
    for(i = 0; i < m; i++) {
        scanf("%d", &avail[i]);
    }

    for(i = 0; i < n; i++) {
        for(j = 0; j < m; j++) {
            need[i][j] = max[i][j] - alloc[i][j];
        }
    }

    int finish[n], safeSeq[n];
    int work[m];

    for(i = 0; i < n; i++)
        finish[i] = 0;

    for(i = 0; i < m; i++)
        work[i] = avail[i];

    int count = 0;
    while(count < n) {
        int found = 0;
        for(i = 0; i < n; i++) {
            if(finish[i] == 0) {
                for(j = 0; j < m; j++) {
                    if(need[i][j] > work[j])
                        break;
                }
                if(j == m) {
                    for(k = 0; k < m; k++)
                        work[k] += alloc[i][k];

                    safeSeq[count++] = i;
                    finish[i] = 1;
                    found = 1;
                }
            }
        }
        if(found == 0) {
            printf("\nSystem is NOT in a safe state.\n");
            return 0;
        }
    }

    printf("\nSystem is in a safe state.\n");
    printf("Safe sequence is: ");

    for(i = 0; i < n; i++) {
        printf("P%d", safeSeq[i]);

        if(i != n - 1)
            printf(" -> ");
    }

    printf("\n");
    printf("raghav N-1wa24cs227");
    return 0;
}

```

Enter number of processes: 5

Enter number of resources: 3

Enter Allocation Matrix:

0 1 0

2 0 0

3 0 2

2 1 1

0 0 2

Enter Maximum Demand Matrix:

7 5 3

3 2 2

9 0 2

2 2 2

4 3 3

Enter Available Resources:

3 3 2

System is in a safe state.

Safe sequence is: P1 -> P3 -> P4 -> P0 -> P2

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